Part A

a) Circuit A

1. $R_T = 20 + 30 + 50$

$= 100\Omega$

$V = IR$

2 $I = V/R = \frac{125}{100} = 1.25\,A$

3 Current remains the same

4 $V = IR$

$20\Omega \rightarrow 1.25 \times 20 = 25\,V$

$30\Omega \rightarrow 1.25 \times 30 = 37.5\,V$

$50\Omega \rightarrow 1.25 \times 50 = 62.5\,V$

5 $P = VI$

$20\Omega \rightarrow 25 \times 1.25 = 31.25\,W$

$30\Omega \rightarrow 37.5 \times 1.25 = 46.88\,W$

$50\Omega \rightarrow 62.5 \times 1.25 = 78.13\,W$

Circuit B

1. $\frac{1}{R_T} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50} = 0.08\Omega = \cancel{0.08}\ 1/0.08 = 12.5\Omega$

2. $I = \frac{V}{R} = \cancel{\frac{125}{0.08}} = \frac{125}{12.5} = 10\,A$

3 $20\Omega \rightarrow 125/20 = 6.25\,A$

$100\Omega \rightarrow 125/100 = 1.25\,A$

$50\Omega \rightarrow 125/50 = 2.5\,A$

4 Voltage across each ~~is~~ ~~th~~ resistor is the same

5 $P = VI$

$20\Omega \rightarrow 125 \times 6.25 = 781.25\,W$

$100\Omega \rightarrow 125 \times 1.25 = 156.25\,W$

$50\Omega \rightarrow 125 \times 2.5 = 312.5\,W$